



**SCHOOL OF ENGINEERING AND NATURAL
SCIENCES**

EEE

Electronics 2

Project

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1. Design Outline and Key Decisions

The objective of this project was to design a two-stage differential-input, single-ended output CMOS operational amplifier using the 65 nm UMC65LL process with the following target specifications: supply voltage of 2.5 V, minimum DC gain of 74 dB, minimum unity-gain bandwidth (UGB) of 10 MHz, maximum overshoot of 5%, minimum slew rate of 5 V/ μ s, load capacitance of 100 pF, and total current consumption below 2000 μ A.

1.1 Topology Selection

A **two-stage Miller-compensated CMOS op-amp** with a PMOS differential input pair and NMOS active load was selected. The first stage converts the differential input signal to a single-ended output and provides the initial voltage gain. The second stage is a common-source amplifier that increases the overall DC gain and drives the 100 pF load capacitor.

The two-stage structure provides a good balance between gain, output swing, stability, and design simplicity, making it the most appropriate choice for this application.

1.2 Key Design Decisions

Bias Current: The second-stage current was determined primarily by the slew-rate requirement. With $C_L = 100$ pF and $SR \geq 5$ V/ μ s, the minimum current needed is $I_{min} = SR \times C_L = 500$ μ A. The final second-stage current was set to approximately 693 μ A to provide adequate margin. The differential pair was biased at approximately 98 μ A per side, ensuring sufficient transconductance for the UGB requirement while keeping total current well below 2 mA.

Transistor Sizing: The channel length was set to $L = 1$ μ m (longer than minimum) to increase output resistance (r_o) and improve DC gain. Initial designs with shorter channels produced insufficient gain. Overdrive voltages were maintained between 0.27 V and 0.28 V, within the required 0.15–0.35 V range. Wide transistors were implemented using the multiplier parameter m , since the process model imposes a maximum single-transistor width.

Frequency Compensation: A Miller compensation capacitor $C_C = 7.5$ pF was placed between the first-stage output and the final output node. A series zero-cancellation resistor $R = 10$ k Ω was added in series with C_C to eliminate the right-half-plane zero, which significantly reduced overshoot from an unacceptable level to approximately 4.877%, just below the 5% limit. A smaller C_C gave better bandwidth but excessive overshoot; a larger C_C reduced overshoot but at the cost of bandwidth.

PMOS input pair: PMOS transistors (PM0, PM1) were chosen for the differential pair because they provide a wider input common-mode range from VSS upward and have lower flicker noise compared to NMOS at the same bias current in this process.

2. Final Design Schematic and Device Parameters

The figure below shows the final two-stage CMOS op-amp schematic with DC operating point values annotated. The design uses a PMOS differential input pair (PM0, PM1), NMOS current mirror active load (NM0, NM1), a PMOS tail current source (PM4), a second-stage NMOS common-source transistor (NM2), and a PMOS current-source load (PM2). The bias reference mirror uses PM3. Miller compensation is implemented with $C_C = 7.5$ pF and $R_Z = 10$ k Ω .

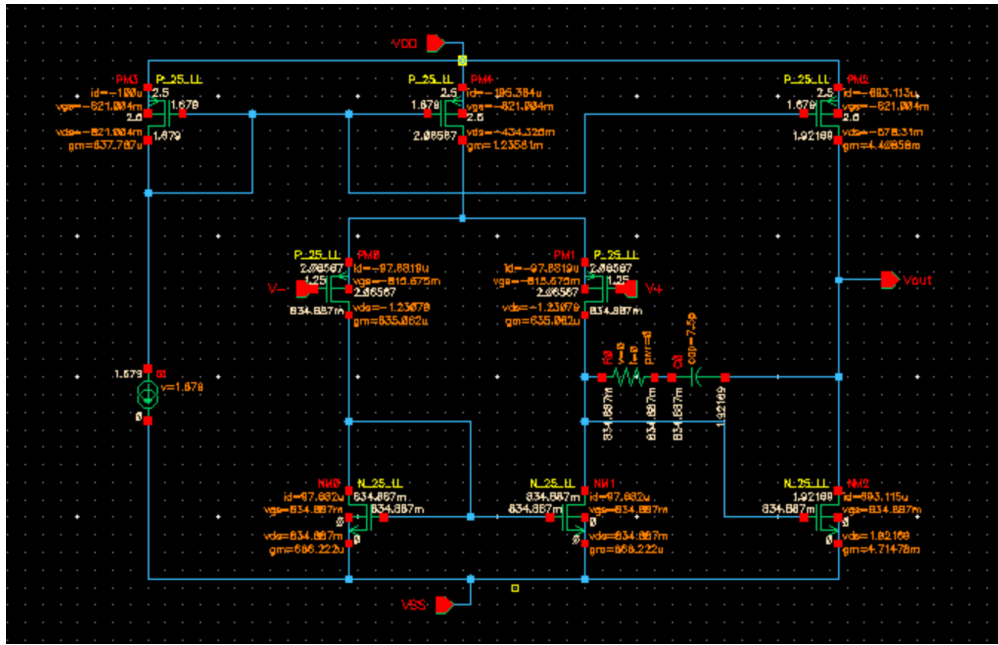


Figure 1. Final two-stage CMOS op-amp schematic with DC operating point values. $V_{DD} = 2.5\text{ V}$, $V_{SS} = \text{GND}$. Total current consumption $\approx 998.479\text{ }\mu\text{A}$. All transistors operate in saturation.

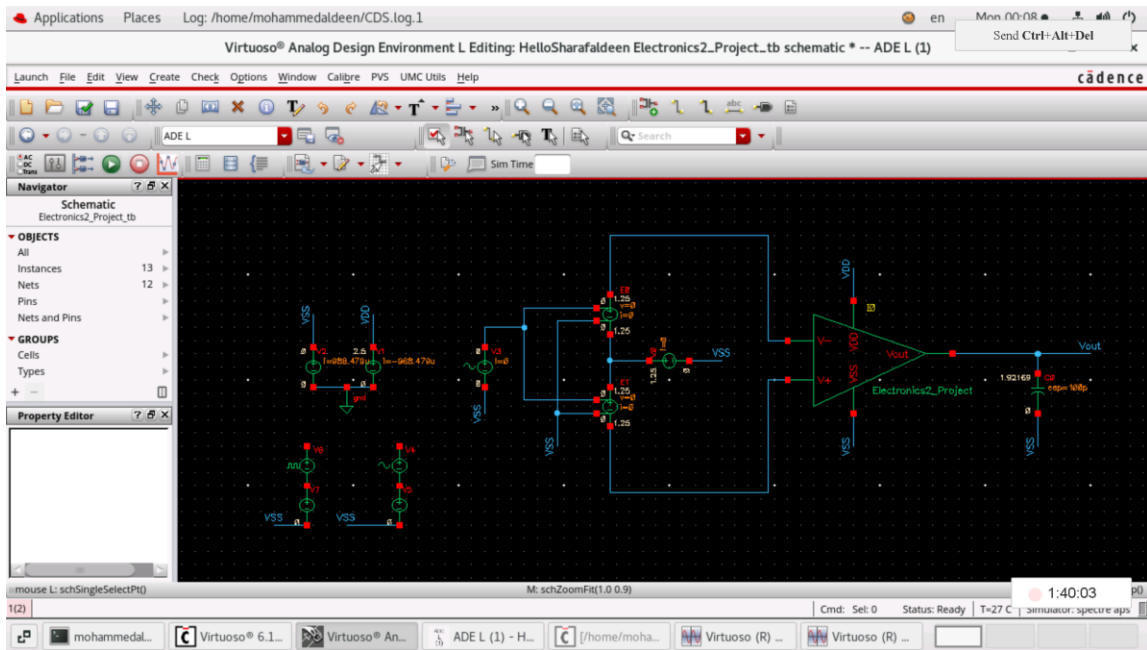


Figure 2. Testbench schematic used for all simulations.

Table 1: Device Sizes

Device	Type	Function	W (μm)	L (μm)	Multiplier (m)	Effective W (μm)
PM3	PMOS (P_25_LL)	Bias / reference mirror	50	1	1	50
PM4	PMOS (P_25_LL)	Tail current mirror	50	1	2	100
PM0	PMOS (P_25_LL)	Input transistor (V $-$)	50	1	1	50
PM1	PMOS (P_25_LL)	Input transistor (V $+$)	50	1	1	50
NM0	NMOS (N_25_LL)	Active load / current mirror	16.8	1	1	16.8
NM1	NMOS (N_25_LL)	Active load / current mirror	16.8	1	1	16.8
NM2	NMOS (N_25_LL)	Second-stage gain transistor	29	1	4	116
PM2	PMOS (P_25_LL)	Second-stage current-source load	35	1	10	350

Table 2: DC Operating Point — Transconductance, Drain Current, and Overdrive Voltage

Device	Type	I_D (μA)	g_m (μS)	V_{GS} (mV)	V_{DS} (mV)	$V_{OV} = V_{gt} $ (mV)	Region
PM3	PMOS	100.34	837.8	-821.0	-821.0	281.1	Saturation
PM4	PMOS	188.05	1289	-821.0	-927.1	281.1	Saturation
PM0	PMOS	97.68	635.1	-815.7	-1230.8	275.8	Saturation
PM1	PMOS	97.68	635.1	-815.7	-1230.8	275.8	Saturation
NM0	NMOS	97.68	666.2	834.9	834.9	271.4	Saturation
NM1	NMOS	97.68	666.2	834.9	834.9	271.4	Saturation
NM2	NMOS	693.12	4715	834.9	1921.7	272.0	Saturation
PM2	PMOS	693.11	4409	-821.0	-578.3	281.1	Saturation

All transistors operate in the saturation region, confirmed by $V_{DS} > V_{DSAT}$ for NMOS and $|V_{DS}| > |V_{DSAT}|$ for PMOS. All overdrive voltages V_{OV} are between 0.27 V and 0.28 V, satisfying the required 0.15–0.35 V range. The total current from VDD is approximately 998.479 μA , well below the 2 mA limit.

3. Calculation of Key Design Parameters

- Min Slew-rate current:

$$\longrightarrow SR = \frac{I_{out,min}}{C_L}$$

$$I_{out,min} \geq SR \times C_L = \frac{5V}{\mu s} \times 100pF = 500 \mu A$$

Therefore, the second-stage current must be larger than or equal to $500 \mu A$ for the slew-rate current requirement to be satisfied

- Compensation capacitor estimate:

$$\longrightarrow C_c = \frac{g_{m1}}{2\pi U_{GB_{min}}}$$

where $U_{GB_{min}} = 10 MHz$

and from my final DC operating point

$$\longrightarrow g_{m1} \approx 635 \mu S$$

$$\longrightarrow C_c \leq \frac{635 \mu S}{2\pi(10 MHz)} = 10.1 pF$$

This means C_c should be less than about $10 pF$ if we want U_{GB} above $10 MHz$

$$\longrightarrow (C_c = 7.5 pF \text{ and } R = 10 K\Omega)$$

The compensation capacitor was chosen by tradeoff. A smaller C_c improved bandwidth but increases overshoot and vice versa for a larger C_c . A $10K$ resistor was introduced in

Series with C_c where it was able to decrease the overshoot from about 30% to less than 5% while keeping the bandwidth in the specific region.

- Min Required g_{m1} from UGB :

$$\longrightarrow UGB = \frac{g_{m1}}{2\pi C_c}$$

$$g_{m1,min} \geq 2\pi \times UGB \times C_c = 2\pi(10\text{MHz})(7.5\text{pF}) = 417\mu\text{S}$$

Therefore, the input differential pair must have g_{m1} greater than or equal to $417\mu\text{S}$

- Min Required input transistor current:

$$\longrightarrow g_m = \frac{2I_D}{V_{ov}}$$

$$I_{D,min} \geq \frac{g_m V_{ov}}{2} = \frac{(417\mu\text{S})(0.27\text{V})}{2} \approx 63.6\mu\text{A}$$

- Extracting K_n and K_p :

$$\longrightarrow K = \frac{2I_D}{\frac{W}{L} V_{ov}^2 (1 + \lambda V_{os})}, \quad \lambda = 0.095\text{V}^{-1}$$

For NMOS, using NM0:

$$\longrightarrow K_n = \frac{2(97.68\mu\text{A})}{16.8(0.2714)^2(1 + 0.095(0.8349))} \approx 146\mu\text{A/V}^2$$

$I_D = 97.68\mu\text{A}$ $V_{ov} = 0.2714\text{V}$ $V_{os} = 0.8349\text{V}$ $\frac{W}{L} = 16.8$
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For PMOS, using PM0:

$$\rightarrow K_p = \frac{2(97.68 \mu A)}{50(0.2758)^2(1+0.045(1.2308))} \\ \approx 46 \mu A/V^2$$

$$\begin{aligned} I_D &= 97.68 \mu A \\ V_{ov} &= 0.2758 V \\ V_{os} &= 1.2308 V \\ \frac{W}{L} &= 50 \end{aligned}$$

• Calculating W/L for each transistor group:

$$\rightarrow \frac{W}{L} = \frac{2I_D}{KV_{ov}^2(1+\lambda V_{os})}$$

$L = 1 \mu m$ was selected because a longer channel length increases r_o , which increases the DC gain. This helped the design reach the required gain.

* For PM0, PM1 (Input Pair):

$$\frac{W}{L} = \frac{2(97.68 \mu A)}{(46 \mu)(0.2758)^2(1+0.045(1.2308))} \\ \approx 50$$

Selected:

$$W = 50 \mu, L = 1 \mu, m = 1$$

$$\begin{aligned} I_D &= 97.68 \mu A \\ V_{ov} &= 0.2758 V \\ V_{os} &= 1.2308 V \\ \frac{W}{L} &= 50 \end{aligned}$$

* For NM0, NM1 (Active Load):

$$\frac{W}{L} = \frac{2(97.68 \mu A)}{(146 \mu)(0.2714)^2(1+0.045(0.8349))} \\ \approx 16.8$$

Selected:

$$W = 16.8 \mu, L = 1 \mu, m = 1$$

$$\begin{aligned} I_D &= 97.68 \mu A \\ V_{ov} &= 0.2714 V \\ V_{os} &= 0.8349 V \\ \frac{W}{L} &= 16.8 \end{aligned}$$

* For NM2 (second-stage transistor):

$$\frac{W}{L} = \frac{2(693.12 \mu A)}{(146 \mu)(0.272)^2(1+0.045(1.9217))}$$

$$\approx 110$$

Selected:

$$W = 29, L = 1 \mu, m = 4$$

$$I_D = 693.12 \mu A$$

$$V_{OV} = 0.272 V$$

$$V_{OS} = 1.9217 V$$

$$\frac{W}{L} = 116$$

* For PM2 (second-stage current-src Load):

$$\frac{W}{L} = \frac{2(693.12 \mu A)}{(146 \mu)(0.2811)^2(1+0.045(0.5783))}$$

$$\approx 360$$

Selected:

$$W = 35, L = 1 \mu, m = 10$$

$$I_D = 693.12 \mu A$$

$$V_{OV} = 0.2811 V$$

$$V_{OS} = 0.5783 V$$

$$\frac{W}{L} = 350 \mu$$

* For PM3 (Bias mirror transistor):

$$\frac{W}{L} = \frac{2(100.34 \mu A)}{(146 \mu)(0.2811)^2(1+0.045(0.821))}$$

$$\approx 51$$

Selected:

$$W = 50, L = 1 \mu, m = 1$$

$$I_D = 100.34 \mu A$$

$$V_{OV} = 0.2811 V$$

$$V_{OS} = 0.821 V$$

$$\frac{W}{L} = 50 \mu$$

* For PM4 (Bias mirror transistor):

$$\frac{W}{L} = \frac{2(188.05 \mu A)}{(146 \mu)(0.2811)^2(1+0.045(0.927))}$$

$$\approx 95$$

Selected:

$$W = 50, L = 1 \mu, m = 2$$

$$I_D = 188.05 \mu A$$

$$V_{OV} = 0.2811 V$$

$$V_{OS} = 0.927 V$$

$$\frac{W}{L} = 100 \mu$$

3.8 Summary of Key Parameters

Parameter	Design Value	Requirement	Status
C_C	7.5 pF	—	—
R_Z	10 k Ω	—	—
I_{SS} (diff. pair tail)	$\approx 196 \mu A$	—	—
I_{D2} (2nd stage)	693 μA	$\geq 500 \mu A$	✓
g_{m1} (PM0/PM1)	635.1 μS	$\geq 471.2 \mu S$	✓
g_{m6} (NM2)	4715 μS	—	—
V_{OV} all devices	0.272–0.281 V	0.15–0.35 V	✓
Total current	998.479 μA	$\leq 2000 \mu A$	✓

4. Verification Simulations

All simulations were performed at nominal process corner, $T = 27^{\circ}\text{C}$, $V_{DD} = 2.5\text{ V}$, with $C_L = 100\text{ pF}$. The op-amp was tested using the testbenches specified in the project description.

Simulation 1 — DC Operating Point: Node Voltages and Device Parameters

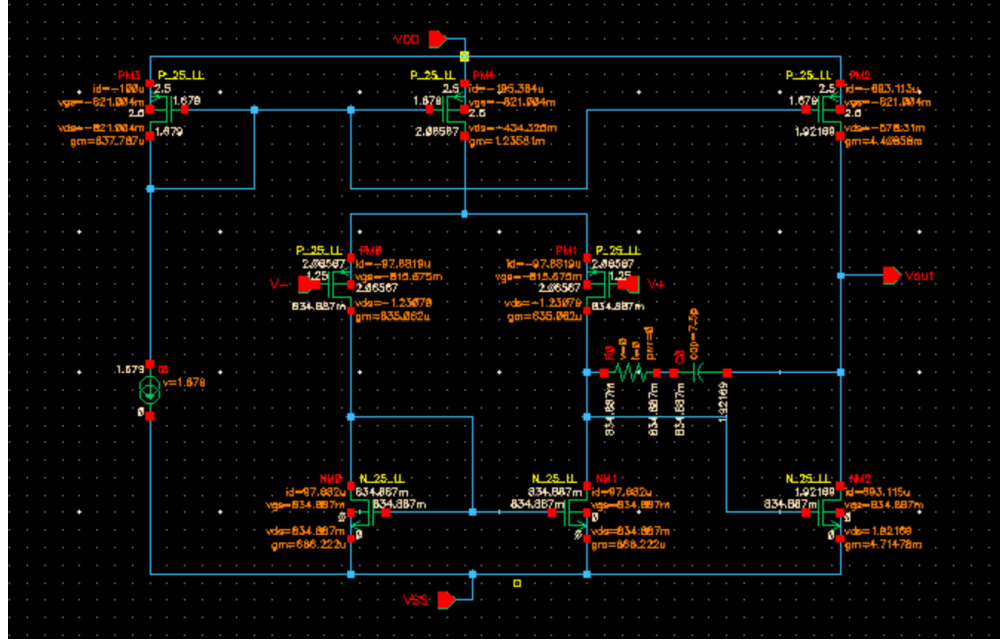


Figure 3. DC operating point simulation showing annotated node voltages and device parameters.

Item	Value
V_{DD}	2.5 V
Bias reference node voltage	$\approx 1.679\text{ V}$
$V_{out,DC}$ (open-loop)	$\approx 1.9219\text{ V}$
C_c	7.5 pF
R_Z	10 k Ω
C_L	100 pF

Total current from V_{DD}	$\approx 998.479 \mu\text{A}$
All transistors in saturation?	Yes — $V_{DS} > V_{DSAT}$ for all devices
V_{OV} range	$0.272 \text{ V} - 0.281 \text{ V}$ (within $0.15\text{--}0.35 \text{ V}$) ✓

Simulation 2 — Open-Loop AC Analysis: DC Gain and Unity-Gain Bandwidth

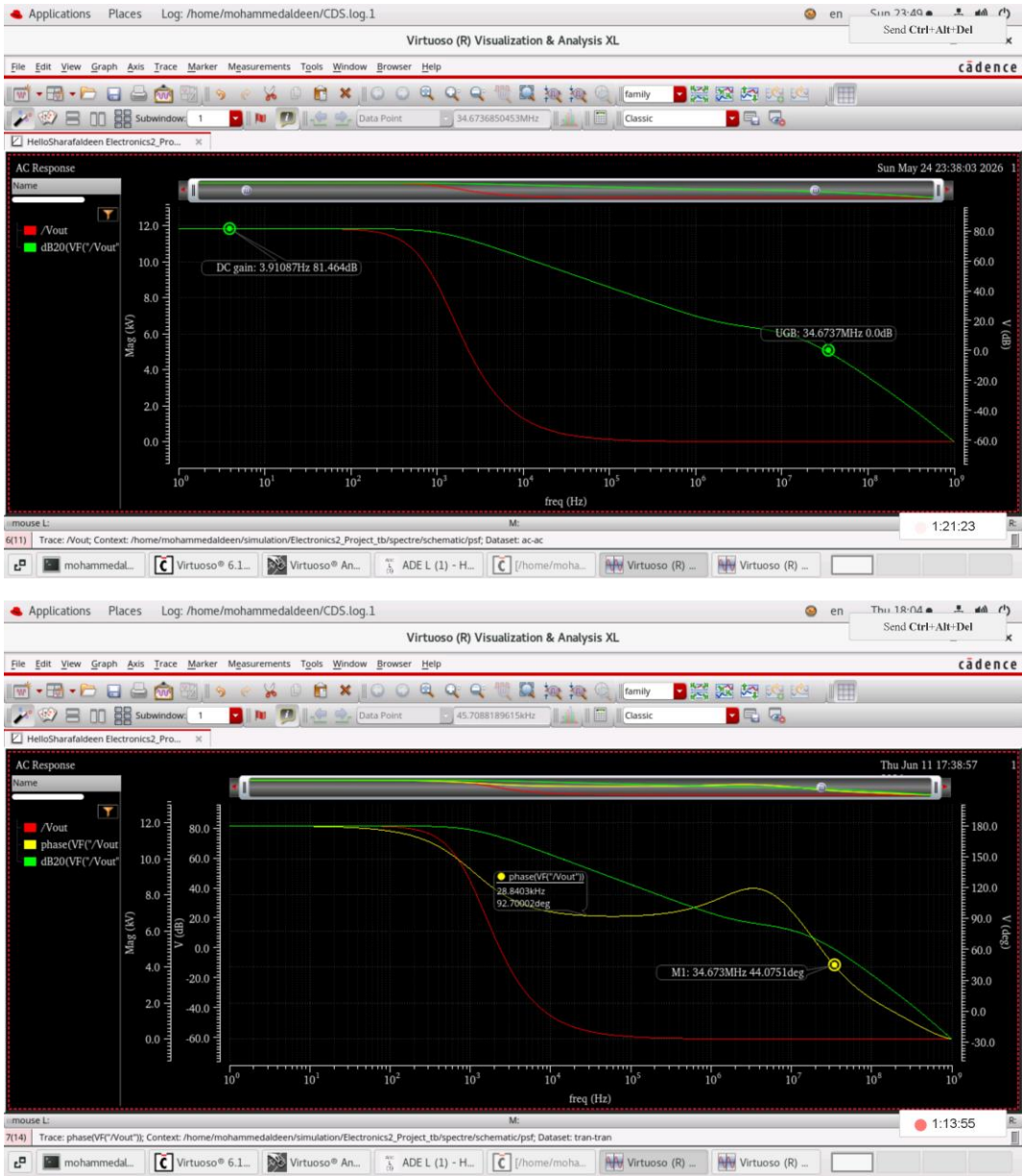


Figure 4. Open-loop AC response. DC gain = 81.44 dB, UGB = 34.673 MHz, PM = 44.1°

Parameter	Measured Value	Requirement	Status
DC Gain	81.44 dB	> 74 dB	PASS ✓
Unity-Gain Bandwidth	34.673 MHz	> 10 MHz	PASS ✓

The open-loop AC simulation confirms that the amplifier achieves a DC gain of 81.44 dB and a unity-gain bandwidth of 34.673 MHz. Both values substantially exceed the minimum requirements of 74 dB and 10 MHz respectively.

Simulation 3 — Transient Response: 1 MHz, 100 mVpp Sinusoidal Input

The amplifier was tested in unity-gain feedback configuration with a 1 MHz sinusoidal input signal, 100 mV peak-to-peak, centered at $V_{\text{bias}} = V_{\text{DD}}/2 = 1.25 \text{ V}$.

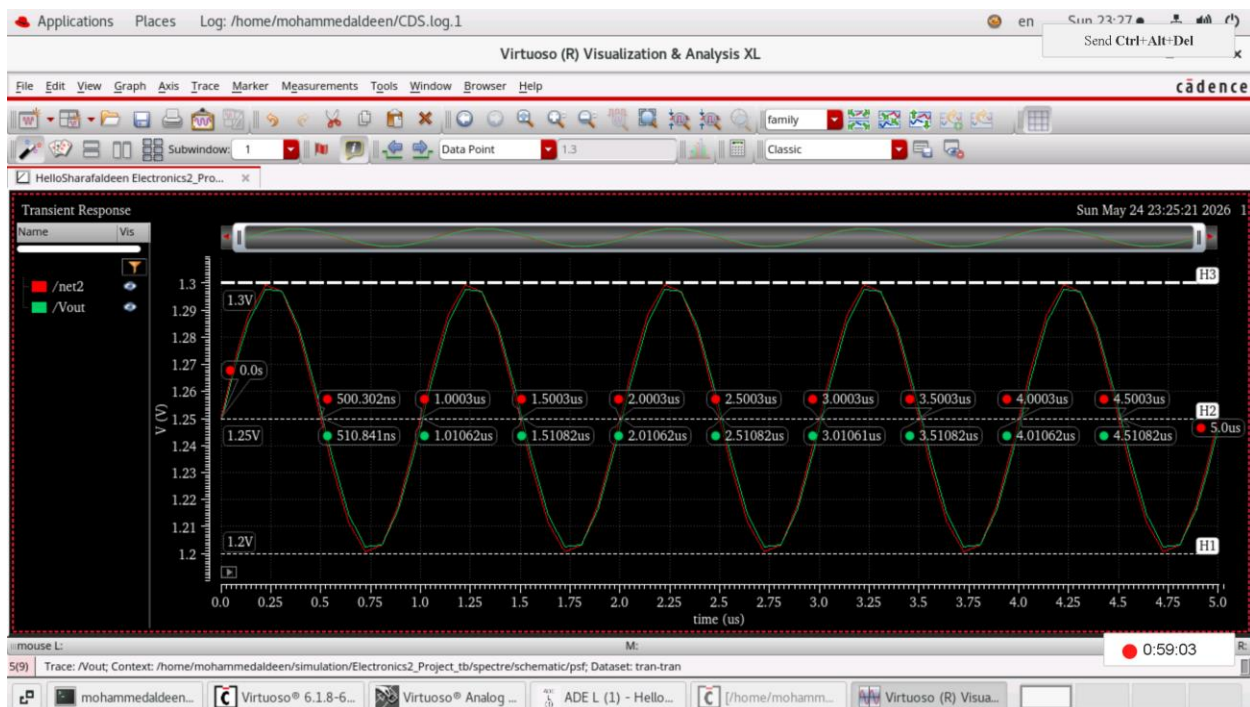


Figure 5. Transient response to 1 MHz, 100 mVpp sinusoidal input. Input centered at 1.25 V, output swings from $\approx 1.20 \text{ V}$ to $\approx 1.30 \text{ V}$ with no visible distortion.

The output follows the input waveform correctly with no visible distortion, confirming proper biasing and stable linear operation at 1 MHz. The amplitude and phase consistency confirm that the amplifier satisfies the bandwidth and stability requirements for sinusoidal signals in the MHz range.

Simulation 4 — Maximum Overshoot: 100 kHz, 100 mVpp Square-Wave Input

The amplifier was tested in unity-gain feedback with a 100 mVpp square-wave input at 100 kHz, centered at $V_{\text{bias}} = 1.25 \text{ V}$ ($V_{\text{DD}}/2$), to measure the maximum output overshoot.

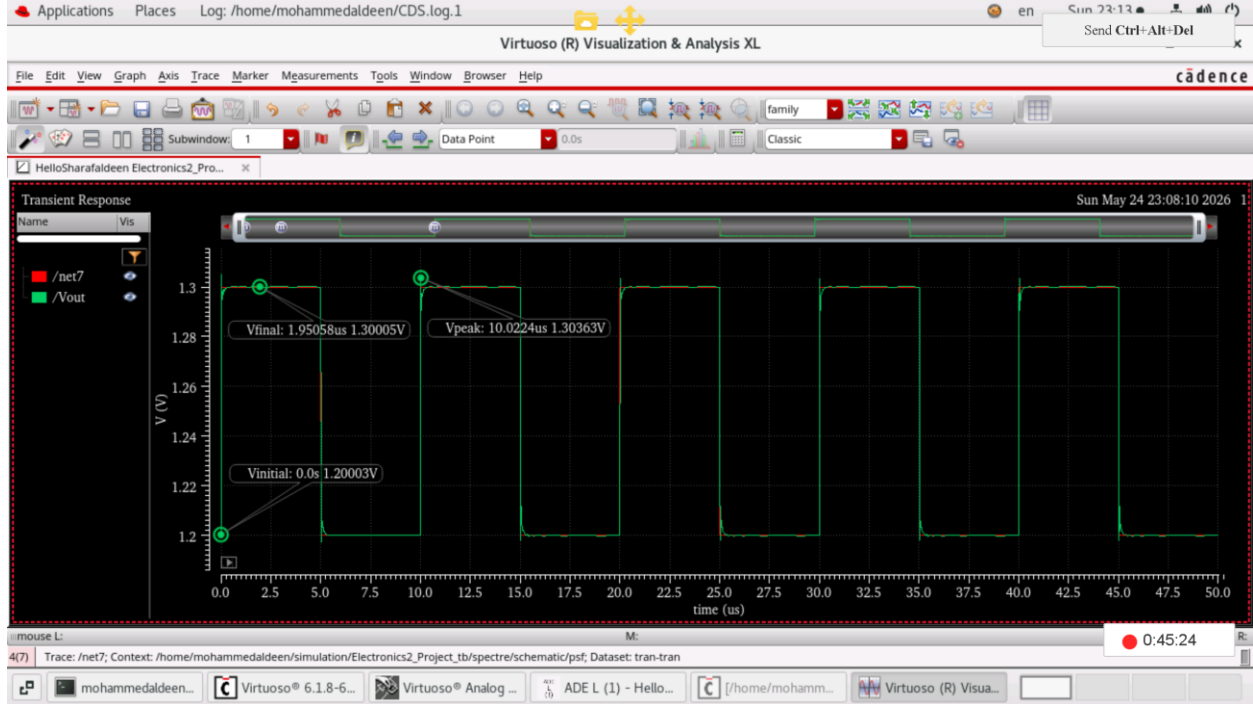


Figure 6. Overshoot simulation: 100 kHz, 100 mVpp square-wave input. $V_{\text{peak}} = 1.304877 \text{ V}$, $V_{\text{final}} = 1.300 \text{ V}$.

The overshoot is calculated as:

$$\%OS = (V_{\text{peak}} - V_{\text{final}}) / (V_{\text{final}} - V_{\text{initial}}) \times 100$$

$$\%OS = (1.304877 - 1.300) / (1.300 - 1.200) \times 100 = 0.004877 / 0.100 \times 100 = 4.877\%$$

Parameter	Measured Value	Requirement	Status
Maximum Overshoot	4.877%	< 5%	PASS ✓

The overshoot of 4.877% is below the required 5% maximum. The zero-cancellation resistor $R_Z = 10 \text{ k}\Omega$ was critical to achieving this result, as designs without R_Z produced overshoot exceeding 5%.

Simulation 5 — Slew Rate: 1 MHz, 1 Vpp Square-Wave Input

The slew rate was measured by applying a 1 MHz, 1 Vpp square-wave input signal in unity-gain feedback. The rising edge of the output was used to determine the slew rate.

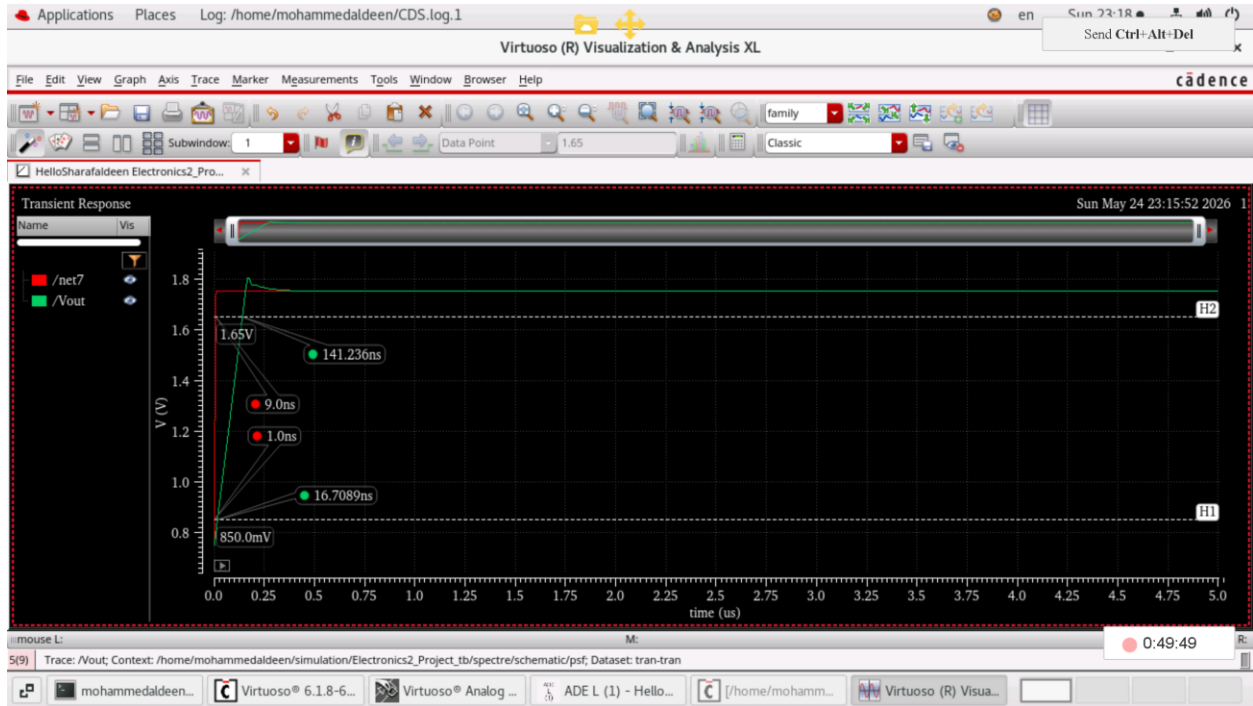


Figure 7. Slew rate simulation: 1 MHz, 1 Vpp square-wave input. Rising edge measured from 0.85 V to 1.65 V.

The slew rate is calculated from the output rising edge:

$$\begin{aligned}
 V_1 &= 0.85 \text{ V}, t_1 = 16.7089 \text{ ns} \\
 V_2 &= 1.65 \text{ V}, t_2 = 141.236 \text{ ns} \\
 SR &= (V_2 - V_1) / (t_2 - t_1) = 0.8 \text{ V} / 124.527 \text{ ns} = \mathbf{6.42 \text{ V}/\mu\text{s}}
 \end{aligned}$$

Parameter	Measured Value	Requirement	Status
Slew Rate	6.42 V/ μ s	$\geq 5 \text{ V}/\mu\text{s}$	PASS ✓

5. Summary of All Specifications

Specification	Requirement	Achieved	Status
Supply Voltage	2.5 V	2.5 V	PASS ✓
Process	65 nm CMOS (umc65ll)	N_25_LL / P_25_LL	PASS ✓
Temperature	27°C	27°C	PASS ✓
Max Total Current	$\leq 2000 \mu\text{A}$	$\approx 998.5 \mu\text{A}$	PASS ✓
Min DC Gain	$\geq 74 \text{ dB}$	81.44 dB	PASS ✓
Min Unity-Gain Bandwidth	$\geq 10 \text{ MHz}$	34.673 MHz	PASS ✓
Maximum Overshoot	$\leq 5\%$	4.877%	PASS ✓
Min Slew Rate	$\geq 5 \text{ V}/\mu\text{s}$	6.42 V/ μs	PASS ✓
Load Capacitance	100 pF	100 pF	PASS ✓
Transistor V_{OV}	0.15–0.35 V	0.272–0.281 V	PASS ✓

All ten project specifications are satisfied. The design achieves a DC gain of 81.44 dB (7.44 dB above minimum), a unity-gain bandwidth of 34.673 MHz (more than $3\times$ the minimum), an overshoot of 4.877% (just below the 5% limit), a slew rate of 6.42 V/ μs (28% above minimum), and a total current consumption of approximately 998.5 μA (50% below the maximum). The two-stage Miller-compensated topology with R_Z – C_C compensation proved to be the optimal choice for meeting all specifications with a simple and area-efficient circuit.